INTRODUCTION TO WEB DEVELOPMENT

Learning Groups: P107, P108, E101, P102

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Lesson 1 learning topics

- **Web development introduction/history of HTML**
- **HTML Basics**
- **Elements and Attributes types**

Lesson 1 Objectives

i. **To understand the principles of creating an effect web page**
ii. **To Learn language of the web HTML & CSS**
iii. **Use specified markup language to develop a web page containing basic HTML tags.**

What Is Web Development?

- Web development refers to building website and deploying on the web. Web development requires use of scripting languages both at the server end as well as at client end.
- Every web developer must have a basic understanding of *HTML, CSS, and JavaScript.*

Web development can be broken down into three layers:

i. **client-side coding** **(front-end),** front end deals with the system interface
ii. **server-side coding** **(backend),** usually behind the scenes dealing with the servers, database and other applications that deliver information to the user.

iii. **Full stack development - includes both front-end and back-end programming. therefore, it represents the complete web development package.**

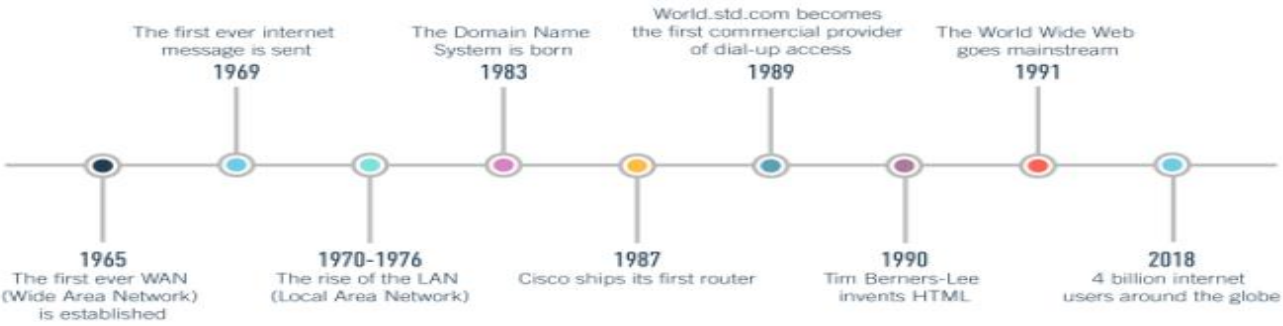
Differences between client side scripting and server side scripting

Client Side Scripting	Server Side Scripting
Source code is visible to user.	Source code is not visible to user because it's output of server side is a HTML page.
It usually depends on browser and it's version.	In this any server side technology can be use and it does not depend on client.
It runs on user's computer.	It runs on web server.
There are many advantages link with this like faster. response times, a more interactive application.	The primary advantage is its ability to highly customize, response requirements, access rights based on user.
It does not provide security for data.	It provides more security for data.
It is a technique use in web development in which scripts runs on clients browser.	It is a technique that uses scripts on web server to produce a response that is customized for each client's request.
HTML, CSS and JavaScript are used.	PHP, Python, Java, Ruby are used.

The difference between web development and web design?

web designer	web development
Web design is an extremely broad field, and will often be broken down into more specific roles such as User Experience Design, User Interface Design and Information Architecture. a *web designer is the architect*	web developer's job to take this design and develop it into a live, fully functional website. *web developer is the builder or engineer.*

A Brief History of the World Wide Web



- Throughout the 1980s and early 90s the Internet was used primarily by large corporations and universities for e-mail and document transfer.
- However, it wasn't until the development of Web browsers and a language called HTML that things really took off.
- HTML (Hyper Text Markup Language) was developed by Tim Berners-Lee while he worked as a researcher in Switzerland. *The original intent of HTML* was to provide a quick and easy way to facilitate information transfer among scientists.
- The basic idea behind HTML is the insertion of tags (for example <body>) within plain ASCII text and the interpretation of those tags by a software program called a browser. (software application for accessing information on the world wide web)

How Does the Web Work?

- The basic principle behind the Web is simple. Programs known as Web browsers (such as chrome, Mozilla Firefox and Microsoft Internet Explorer) located on one computer request Web pages from Web servers.

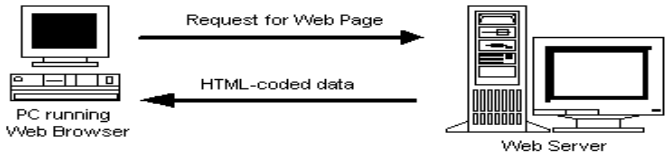


Figure A-1: A Web Browser Talking to a Web Server

Web browsers not only let you view Web pages but also have built-in support for Email, connecting to Usenet discussion groups, and downloading load files via ftp (file transfer protocol). The secret to all of this is the URL

Web Servers: - Web servers are software applications running on Web sites that handle requests from Web browsers

Web server software returns Web pages (including pictures, audio and possibly video), encoded in HTML, back to their Web browser clients.

The Uniform Resource Locator (URL) *–is the web address* Computers on the Internet use the URL to make a connection between a Web browser and the target Web server.

URL contains

- **Protocol -**rules that govern how computers communicate to each other) **i.e.** http
- **Host or domain:** www.html.com
- **Path of the resource: full –form –** http://www.html.com
  Sections for Http url



http:// or https://

The "http" stands for Hypertext Transfer Protocol. It lets the browser to know which protocol it is going to use to access the information specified in the domain. An "https" protocol is short for "Hypertext Transfer Protocol Secure" and indicates that information transmitted over HTTP is encrypted and secure. After the http or https is the colon (:) and two forward slashes (//) that separate the protocol from the remainder of the URL.

www.

www stands for world wide web and is used to distinguish the content. This portion of URL is not required and many times can be left out. For example, typing "http://computerhope.com" would still get you to the computer hope website. This portion of the address can also be substituted for an important subpage known as a subdomain

computerhope.com

"computerhope.com" is the domain name for the website. The last portion of the domain is known as the domain suffix, or TLD. It is used to identify the type or location of the website. For example, ".com" is short for commercial, ".ac" for academic other extensions

Entity	Extension	Full Web Site Name
Corporate	com	http://www.merant.com
Government	gov	http://www.whitehouse.gov
Educational	edu	http://www.seas.smu.edu
UK company	co.uk	http://www.merant.co.uk

/jargon/u/

Next, "jargon" and "u" are the directories where the web page is on the server. In this example, the web page is two directories deep. To find the file on the server, it would be in the */public_html/jargon/u* directory. With most servers, the public_html directory is the default directory containing the HTML files.

url.htm

Finally, url.htm is the actual web page on the domain you're viewing. The trailing .htm is the file extension of the web page that indicates the file is an HTML file. Other common file extensions on the Internet include .html, .php, .asp, .cgi, .xml, .jpg, and .gif. Each of these file extensions performs a different function, like all the different types of files on your computer.

TOOLS FOR CREATING HTML DOCUMENTS

HTML tool editors - converts text and layout interface input into actual HTML code

- There are so many software packages available to develop HTML. The software packages can be grouped into two main categories:

i. text-based (or code-based) editors i.e. (notepad)

ii. WYSIWYG (*what you see is what you get*) editors i.e. (dream weaver)

List of web Editors

1. Atom · 2. Visual Studio Code · 3. Brackets · 4. Sublime Text 5. Light Table 6. Notepad/notepad++ etc.

Use any text editor provided you are conversant with it

Class work notepad /sublime text

Introduction to HTML and Types of Tags

HTML- is a language to format web pages

- **the main markup language for describing the structure of web pages**

HTML as described earlier is a markup language not a programming language, like Java, Ruby, PHP, etc. You need a web browser to view the HTML pages. The web browsers do not display the HTML tags, but uses the tags to interpret the content of the web pages.

HTML stands for Hypertext Markup language

Hyper – (link) allows us to navigate between pages or within same page. www.html.com

Text – the actual information/ content to be displayed on the web page

Markup Language - a tag based language used to format web pages understood by browsers **like** chrome, Firefox, safari, opera etc.

HTML is built by following or by referring SGML'S standard set of specifications, hence we say *HTML is evolved from SGML*. *(standard Generalized markup language)*

SGML is a standard set of specifications followed for specifying or building any markup language.

- **Website:** is a collection of inter-linked related webpages.
- **Webpage**: is a well formatted document used to share information over the internet.
- **Browser:** is a software for interpreting and displaying HTML files (Firefox, chrome, opera, safari)

Some other popular markup languages are DHTML (Dynamic mark-up language), XHTML (Extended Html), XML (Extensible Mark-up language)

Html Versions

- **HTML 2.0** : Some of the Netscape/Microsoft extensions, and did not support tables, or ALIGN attributes.
- **HTML 3.0** : Upgrade the features and utility of HTML. However, it was never completed or implemented.
- **HTML 3.2** : Support for TABLES, image, heading and other element ALIGN attributes, and a few other finicky details.
- **HTML 4** : It includes support for most of the proprietary extensions, plus support for extra features (Internationalized documents, support for Cascading Style Sheets, extra TABLE, FORM, and JavaScript enhancements).
- **HTML 5** : The <canvas> element for 2D drawing,<video>,<audio> & some extra features support like New form controls, like calendar, date, time, email, url, search.

Version	Year
HTML	1991
HTML+	1993
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML 1.0	2000
HTML5	2012
XHTML5	2013

HTML5 provides a faster and more robust approach to web development.

Advantages of HTML	Disadvantages Of HTML
Html is easy to learn and understand	**It cannot be used to develop a dynamic website**
Html is supported by all browsers	**There is no complete acceptable standard of HTML**
Html is easy simple to edit	**There are many incompatibilities of HTML**
Display changes instantly	**It is complex to design attractive web page only using HTML .**
Html can integrate with other browsers easily	**Html has no programing capabilities**

Html Tags

HTML tag is a command that provides the directions for visual content that one sees on the web.

Html is bound by angular brackets (<>) that always opens with a < (less than) sign and closes with a > (greater than) sign. **It controls the appearance, layout and flow of the web page.**

A tag contains three parts: element (identification of tag), attribute and value.

HTML tags can be of two types:

i. Paired Tags
ii. Singular Tags

1. Paired Tags

It is also called container tag. Have their companion tags opening/starting and closing/ending tags.

Tag Examples

- **html> ... </html>** — The root element. ...
- **<head>** ... **</head>** — The document **head**. ...
- **<title>** ... **</title>** — The page **title**. ...
- **<body>** ... **</body>** — The page's content. ...
- **<h1> ... </h1>** — A section heading. ...

- ** … ** — A paragraph. …
- **<a> … ** — A link. …
- **<img>** — An image. The **img** element lets you insert images into your web pages

2. **Unpaired tags** *singular, void tags or standalone tags*.

Examples

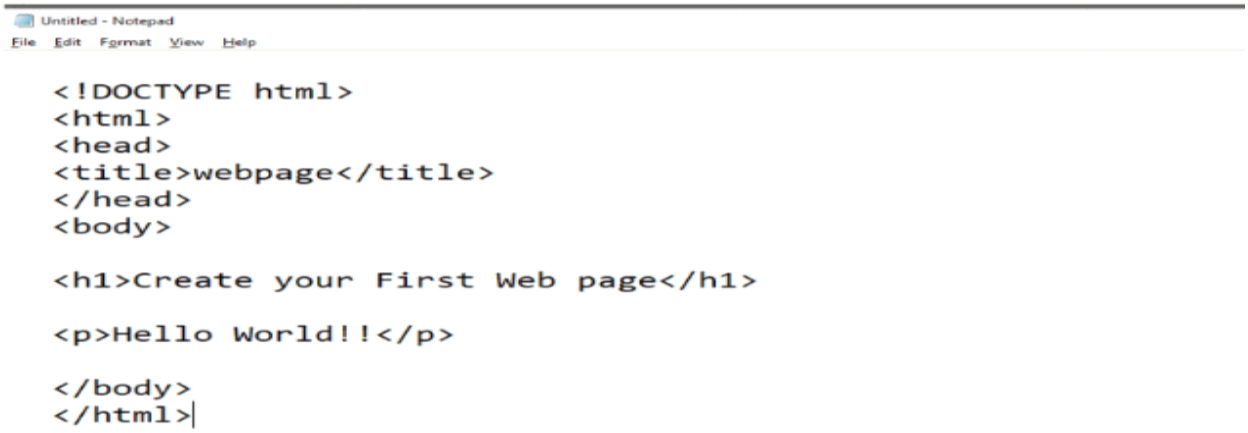
**
** insert a link break.

<hr/> defines a horizontal rule.

Basic structure of html document

Step 1: Open Notepad (Windows)

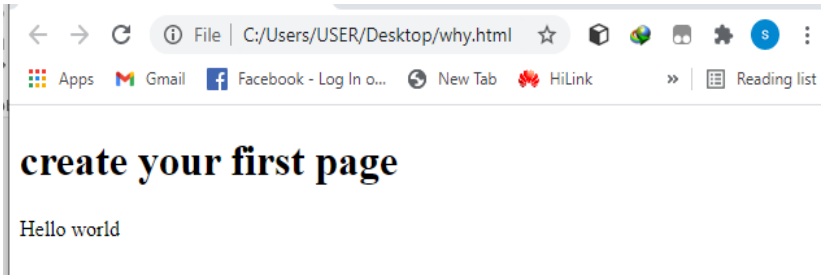
Step 2: Write code in HTML



Step 3: Save the HTML file with .htm or .html extension.

Step 4: Open the HTML page in your web browser. i.e. chrome, fire fox Mozilla

Output



Building blocks of HTML

- **Tags:** An HTML tag surrounds the content and apply meaning to it. It is written between < and > brackets.
- **Attribute: An attribute is used to *define the characteristics of an HTML element* and is placed inside the element's opening tag.**

  All attributes are made up of two parts – a name and a value

  i. **Name -** The name is the property you want to set
  ii. **Value -** The value is what you want the value of the property to be set and always put within quotations

Example of HTML attribute

Name: example <p align

Value: example = "left "

```
<!DOCTYPE html>
<html>

   <head>
      <title>Align Attribute  Example</title>
   </head>

   <body>
      <p align = "left">This is left aligned</p>
      <p align = "center">This is center aligned</p>
      <p align = "right">This is right aligned</p>
   </body>

</html>
```

Core types of Attributes The four core attributes that can be used on the majority of HTML elements (although not all) are:

- **Id**
- **Title**
- **Class**
- **Style**

The style Attribute

The style attribute allows you to specify Cascading Style Sheet (CSS) rules within the element.

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```
<!DOCTYPE html>
<html>
<head>
<title>The style Attribute</title>
</head>
<body>
<p style="font-family:arial; color:#FF0000;">Some text...</p>
</body>
</html>
```